

Nkosiyezwa Malibongwe Mpanza

Software Engineer

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Portfolio: nkosiyezwa-portfolio.vercel.app

Portfolio: nkosi-mpanza-dev.higgsfield.app

PROFILE

Software Engineer with professional experience developing backend applications using Python, Django, REST APIs and PostgreSQL, with academic experience in Java. Diploma-qualified in Computer Science from Tshwane University of Technology (TUT), focused on building secure, reliable and maintainable software solutions.

EXPERIENCE

Intern Software Engineer — Aestus Analytics Pty Ltd

October 2024 – January 2026

- Developed backend applications using Python and Django framework.
- Worked on the Secure Document Sharing Project for secure document sharing between faculty members and students.
- Developed and integrated RESTful APIs.
- Used PostgreSQL database for data storage and management.
- Applied blockchain and cryptography concepts to support secure data handling.
- Participated in debugging, testing and software development activities.

PROJECTS

Secure Document Sharing System

Developed a secure document sharing platform using Python, Django, REST APIs and PostgreSQL, with Google OAuth authentication and role-based access for faculty, students, and admins. The system enabled secure document exchange with a focus on confidentiality, data integrity, and access logging.

Shopping Cart Java GUI Application

Built a Java Swing desktop application simulating a small e-commerce store, featuring a product catalog, shopping cart management, automatic VAT and discount calculation, shipping options, and save/load cart state between sessions.

SAWeatherApp

Developed a Python application that fetches real-time temperature, humidity, and weather conditions for any city in South Africa using the OpenWeatherMap API, with automatic fallback to geographic coordinates when a city name is not recognized.

EDUCATION

Diploma in Computer Science

Tshwane University of Technology (TUT) – Completed

CERTIFICATIONS

- Oracle Cloud Infrastructure (OCI) AI Foundations Associate
- Oracle Cloud Infrastructure (OCI) Generative AI Professional
- Diploma in Computer Science – Tshwane University of Technology (TUT)

SKILLS

Programming Languages:	Java, Python, Kotlin, JavaScript, SQL/PLSQL, HTML, CSS
Backend & Frameworks:	Django, Java MVC (Servlets, JSP), Java Swing, REST APIs, OpenWeatherMap API
Databases:	PostgreSQL, SQL, PL/SQL
Tools:	Git, GitHub, IntelliJ IDEA, Visual Studio Code, Android Studio, Postman, MySQL Workbench, Jupyter Notebook, Linux, Windows
Software Practices:	Agile Development, Debugging, Testing, Clean Code Principles